



Research paper

Long-term optimization of hydro & solar power electricity generation in the Taoussa area of Mali using the MESSAGE model

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ABSTRACT

Optimization for a long-term electricity supply strategy with renewable energy is essential for electricity security and financial and environmental sustainability. The main goal of this study is to find the equilibrium between electricity demand and sustainable optimal electricity supply mix scenarios of the Taoussa area at the least cost for the development of north in Mali, by using the Model for Energy Supply Strategy Alternatives and their General Environmental Impact (MESSAGE) from the International Atomic Energy Agency (IAEA) for the period 2020–2035. This model is applied to obtain the least-cost electricity and supply of electricity under electricity demand forecasted by reference scenario (RS) and two alternative scenarios (Low scenario, LS; High scenario, HS), technical information on power generation per technology and regional information. The result obtained from the optimization of the power management strategy shows that the final electricity generation is 1.02 times greater than the electricity demand and the annual increase of power generation is equivalent to 8.13%, 10.32% and 12.56%, under different scenarios. The maximum power from PV and hydropower by scenarios are from April to May in the hot season. Thus, the proposed MESSAGE model is useful in improving the economical-environmental optimization of the operation of large-scale hybrid renewable electricity supply.

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1. Introduction

1.1. Background

Currently, all countries, whether developed or developing, are subject to three major constraints which require them to define a strategy for energy change: market tensions and rising hydro-carbon prices, energy security needs in the long term, and climate degradation, with increasing local impacts. The situation is all the more worrying in countries like Mali, which has no conventional energy resources and is dependent on the outside world for its energy supply (AFD/IFAD, 2016).

Mali's energy supplies were 6928 ktoe in 2018. Biomass accounts for nearly 74% of the country's energy supply, followed by petroleum products for 24% and hydroelectricity and solar PV for 2%. All petroleum products for consumption are imported. The total installed capacity of EDM-SA (Energy Company of Mali)'s generating fleet was 719.9 MW in 2018, mainly from hydro-electric sources (25.52%), thermal power (48.76%) and the inter-connection with the Republic of Cote d'Ivoire (25.73%) (UEMOA,

2019). The share of thermal power in the total generation is still quite high and EDM-SA's thermal generation fleet is dilapidated, which leads to an expensive production cost.

However, the contribution of renewable energy to the overall energy balance is still very low compared to the country's potential. Mali's renewable energy potential is considerable and still largely untapped, crossed by the Niger (1700 km) and Senegal (800 km) rivers, the identified hydroelectric potential is 1150 MW on about twenty sites with a corresponding average annual producible of 5600 GWh approximately. Currently, nearly 840 MW of this potential remains available. With a high solar irradiation (5 to 7 kWh/m²/d) well distributed on the territory, the photovoltaic potential is also considerable with a sunshine duration of around 10 h per day. A wind band of 3 to 7 m/s in the Sahelian and Saharan areas of the country (JatroREF, 2014; CERECE/CEDEAO, 2015a,b; SeforALL, 2019).

In this context, the Economic Community of West African States (ECOWAS) has developed the National Renewable Energy Action Plans (PANER) and the United Nations Sustainable Energy for All National Action Program (SE4ALL). These plans and programs for Mali aim to increase the capacity connected to the grid in renewable energies to 977.4 MW or 52.5% by 2030 (including 538 MW of solar, 389 MW of hydro, 20 MW of wind and 30 MW

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Nomenclature

Abbreviations

CS	Cost savings (US\$/kWyr)
E_{lost}	Electricity lost (GWh)
L_{LRF}	Length of load regions, fraction
O&M	Operation and Maintenance
PVGIS	Photovoltaic Geographical Information System
SS	Share by season k
SPD	Share by part of the day
STD	Share by typical day j
T&D	Transmission and Distribution

Parameters

C_e	Energy consumers
D	Day
D_{et}	Exogenous Demand
ND_H	Number of the hour for a typical day j
ND_k	Number of days in the season k
N_H	Number of hours
nT	Number of periods in the model
Pe	Producers of energy carrier

Variables

d_t^o/d_t^c	Discount factors applied for operating and capital cost in period t , respectively
Δ_t	Length of period t in years
ED_t	Demand for electricity in period t (GWh)
EP_t	Production of electricity in period t (GWh)
FC_{qt}	Fix O&M costs of technology q in period t (US\$/kWyr)
HA_D	Hourly load average for each day D
HA_k	Hourly load average in the season k
HD_D	Hourly load average for each day d in week m by season k
HD_{N_H}	Hourly load average of typical day j
H_{ijk}	Hour number during LR i of typical day j in season k
HP_{N_H}	Hourly load average in part p of typical day j
HT_D	Hourly load average for typical day j in week m by season k
IC_{qt}	Investment costs of technology q in period t (US\$/kWyr)
i_{qt}	Specific investment of technology q in period t
ND_j	Number of the day in typical day j

n_j	Number of days of type j in season k
N_{qt}	Conversion efficiencies of energy technologies
O_{qt}	Operating costs of technology q in period t
$VC_{T\&D_t}$	Variable O&M costs of T&D in period t (US\$/kWyr)
VC_{qt}	Variable O&M costs of technology q in period t (US\$/kWyr)
W_{ij}	Electricity consumed during load regions i of typical day j in season k
X_{qt}	Fuel consumption of technology q in period t
Y_{qt}	Capacity variables for annual new installations of technologies

by power plants at all levels of the energy chain. These CO₂ emissions are causing global climate change (Roh and Kim, 2017; Alshammari, 2021; Gambhir et al., 2013). Energy supply plan of Macedonia is based on the construction of new coal-fired power plants and the modest increase in renewable technologies (hydroelectric, wind and solar power plants) to reduce CO₂ emissions while guaranteeing the security country's energy supply (Popov et al., 2015). For power systems, Roh and Kim (2016) have introduced the possibility and assessed the implications of deploying a nuclear power plant in Nigeria's energy mix system using the MESSAGE code for informed forecasting of electric power demand considering a set of criteria such as minimizing the systemic discounted cost under a set of constraints and the various limits imposed on the environmental impacts associated with the use of this technology. Kichonge et al. (2015) carried out a technical feasibility study and a financial study for the proper functioning of an optimal energy mix system adapted to Tanzania to facilitate continuous electricity production. MESSAGE model was used based on the energy data of the Czech Republic and Syria to optimize the expansion and operation of the power system to meet future energy needs at the lowest cost using an appropriate mix of technologies and resources (Rečka, 2011; Hainoun et al., 2010). In Chile, the MESSAGE planning tool is used to study the effect of the generation of electricity using flowing water in the southern part of the country, the atomic power plant and the intermittent effect of green energy (Watts and Martinez, 2010). The above studies demonstrate that the MESSAGE is a reliable regional energy analysis tool. Few studies have focused on Mali's renewable energy potential and planning, and analysis based on MESSAGE could well address this issue.

1.3. Contribution and organization of the paper

In this work, the long-term planning horizons model MESSAGE is used to optimize renewable electricity supply to meet the challenge of satisfying the future electricity demand of the Taoussa area at the least cost-optimal technology mix and lowest CO₂ emissions. The novel contribution of this article is to introduce a method based on the concepts of load regions and load curves (subdivisions of the year, weeks and days) which are essential to represent forms of energy. This is particularly the case for electricity, which must be supplied exactly when it is needed. The supply of electricity must correspond to the demand. But the demand for electricity changes constantly during the day. Suppliers must produce more electricity when demand is high and less when it is low. In other words, suppliers must avoid

of bioelectricity), to contribute to socio-economic development, improve access to energy and protect the environment (JatroREF, 2014; CERECE/CEDEAO, 2015a,b).

1.2. Literature review

Many studies used the optimization model of energy supply systems (MESSAGE) to assess capacity expansion and energy production policies. In Saudi Arabia, China and South Korea, MESSAGE modeling is used to account for the CO₂ emissions caused

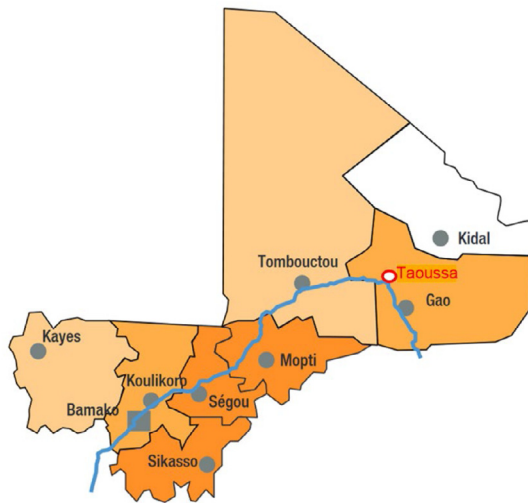


Fig. 1. Location of Taoussa area in Mali (CSAO/OCDE, 2015).

underproduction and overproduction. To take account of this situation MESSAGE offers the possibility of dividing each year into a certain number of time slices which we will call “load regions”. MESSAGE then calculates a load curve from the definition of these load regions and the annual electricity demand distribution over these regions. The Consideration of load curves improves the representation of electricity demands and the uses of different types of equipment to satisfy it.

The remainder of this paper is arranged into the following five sections:

Section 2 gives information on the potential of solar hydro and the historic consumption of electricity in the Taoussa area. The proposed simulation and optimization model for the long-term electricity demand & supply strategy are described in Section 3. In Section 4, input data for the MESSAGE model is presented. In Section 5, optimization results and discussions are presented, and in the end, the conclusions and future academic work are described in Section 6.

2. Energy supply and demand data of the Taoussa area

The Taoussa area is mapped north of Mali between the Gao and Tombouctou region as shown in Fig. 1.

The Taoussa area has a latitude of 16°25'0" North and a longitude of 3°10'60" West. The Taoussa dam is located in the Niger Delta, it is 280 km from Tombouctou upstream and 120 km from Gao downstream, and consists of a 1000 m wide dam in the Niger River valley, with a height of 18 m. As a result, a lake of 1572 km² will have emerged, averaging 300 km in length, 5 km in width and 3 m in depth. In terms of energy, the dam would allow the production of 25 MW of electrical energy to supply the Gao region (GWI, 2010). The population of the Taoussa area was 3.84% of the total population of Mali (19.66 million people) for the year 2019 and its surface represents 13.74% of the surface of Mali (1,241,248 km²).

2.1. Hydroelectric power potential

According to Fig. 2, the hydroelectric power station in Taoussa will be located in the second branch of the river between the spillway and the central embankment. Turbine flows will be returned to the bed of the river directly downstream of the plant, in the central branch of the Niger River. The characteristics retained for the hydroelectric power station in Taoussa are as follows (GWI, 2010):

- Number of groups: 5;
- Nominal installed power per group: 5 MW;
- Total nominal installed power: 25 MW;
- Average annual production: 118 GWh.

2.1.1. Historical evolution of the average monthly flows in the Niger river at Taoussa

Fig. 3 shows the historical data of the flows in the Niger River, which indicates that the Niger wet season in Taoussa starts from July until December–January. The maximum flood occurs during November, and December and exceeds on average 1800 m³/s. The month of April corresponds to the period of low water with average monthly flows of less than 200 m³/s.

As shown in Fig. 4, the three curves start to increase progressively from July and attain their maximum level between November and December during the year.

2.1.2. Monthly energy characteristics

Figs. 5 and 6 illustrate the average monthly energy characteristics guaranteed at 10% and 90% of the time obtained by the release of water from a storage reservoir to supplement downstream flow. With 5 groups in operation, the minimum power and producible guaranteed 90% of the time varies between 4.5 MW (3.3 GWh) in July and 16.3 MW (12.2 GWh) in December, the maximum power and producible guaranteed 10% of the time evolves between 10.8 MW (8 GWh) and 22.5 MW (16.7 GWh) during the year, the average power and producible varies between 6.9 MW (5 GWh) in June and 17.4 MW (13 GWh) in December and then the power and the average producible feasibility ($Q = 5 \times 60 \text{ m}^3/\text{s}$) varies between 6 MW (4.4 GWh) in August and 14.5 MW (10.35 GWh) in November.

2.2. Taoussa's solar potential

The sunshine in the Taoussa site is fairly regular except for certain days of the rainy season and dry cool. Table 1 presents the average monthly solar radiation and temperature.

The average temperature of the cold weather month (January) is 17 °C and the average temperature of the warm weather month (May) is 43 °C from 1991 to 2020. The annual average solar radiation in the period 2014–2016 is 6.81 kWh/m²/day with a duration from 6 a.m to 5 p.m per day. According to this basis, the energy received on the entire territory of Taoussa (170572 km²) is 1161.60 TWh/m²/day. Taoussa's electrical energy requirement for the reference scenario is estimated to be 472.61 GWh in 2035, which can be covered 2457.83 times by one day of solar radiation. This information will be used for the sizing of the photovoltaic systems.

2.3. Historic consumption of electricity during the 2013–2019 period

The northern city of the country like Gao is supplied by a small diesel-fueled thermal power plant. The installed capacity of the Gao thermal power plant is 10.99 MW. In Table 2, the peak load in 2019 was 5.904 MW with an average increase rate of 19.98%. The recorded gross electricity production was 29.935 GWh in 2019. This thermal power generation capacity is economically and environmentally very deficient overall.

3. System modeling (MAED-MESSAGE)

Fig. 7 illustrates the interactions between the bottom-up approach Model for Analysis of Energy Demand (MAED) and the bottom-up optimization MESSAGE model, relating the input and output data of each model.



Fig. 2. Overview of the dam and the immediate area of influence (GWI, 2010).

Table 1

Annual temperature and solar radiation of Taoussa.

Source: VOYAGES (2020) and PVGIS.

		Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sep.	Oct.	Nov.	Dec.
Monthly average temperature (°C) from 1991 to 2020	Minimum	17.00	19.00	23.00	27.00	30.00	30.00	28.00	27.00	27.00	27.00	22.00	18.00
	Maximum	30.00	34.00	38.00	41.00	43.00	42.00	39.00	37.00	39.00	40.00	36.00	32.00
	Average	23.50	26.50	30.50	34.00	36.50	36.00	33.50	32.00	33.00	33.50	29.00	25.00
Hours of sunshine from 1991 to 2020	Daily average	8.50	9.00	8.50	8.50	9.00	7.50	8.00	8.00	8.50	9.00	9.50	8.50
	Total	270.00	260.00	270.00	255.00	275.00	225.00	250.00	250.00	255.00	280.00	280.00	260.00
Monthly solar irradiation (kwh/m ² /month)	2014	207.69	212.81	237.66	231.51	190.49	192.84	197.44	198.07	211.65	224.89	202.22	219.88
	2015	213.05	205.35	227.16	231.89	216.68	192.92	188.71	184.73	194.32	214.31	216.86	211.23
	2016	187.68	215.73	232.60	205.93	211.37	191.31	195.93	192.93	200.39	219.50	208.86	203.77
	Average	202.81	211.30	232.47	223.11	206.18	192.36	194.03	191.91	202.12	219.57	209.31	211.63

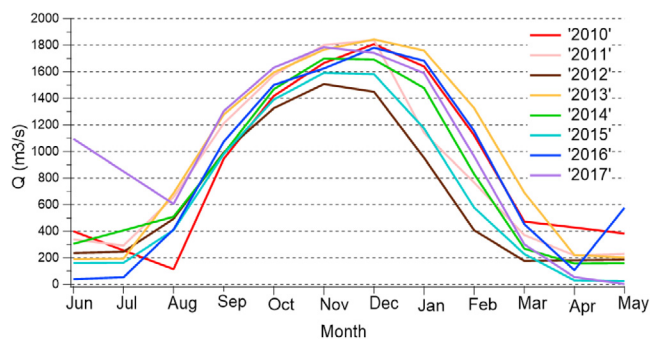


Fig. 3. Comparison between the 8 years of historical data (2010–2017) of the average monthly flows in the Niger River at Taoussa.
Source: National water authority of Mali.

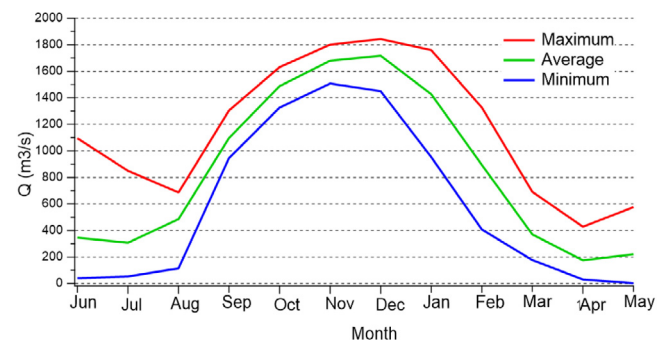


Fig. 4. Extreme flows of the Niger River at Taoussa during the period 2010–2017.
Source: National water authority of Mali.

Table 2

Evolution of electricity production (National Energy Authority of Mali, 2019).

Year	Peak power in MW	Electricity production in GWh
2013	2.16	10.46
2014	2.28	12.07
2015	3.80	17.89
2016	4.53	21.88
2017	4.97	24.01
2018	5.86	27.25
2019	5.90	29.94

The MAED methodology is based on the technique of simulating the long-term energy demand of the country/region's future development scenarios. Each scenario was developed based on the factors of economic development, demographic evolution, lifestyle change and technological change that directly affect energy demand. After that, the demand module (MAED) calculates the total annual final energy demand of each consuming sector, which is the most indispensable input data in the MESSAGE model.

MESSAGE is a system engineering optimization model used for medium- and long-term power system planning, policy on

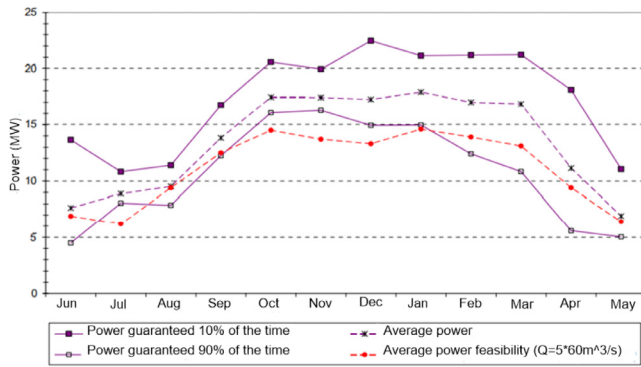


Fig. 5. Monthly power (MW).
Source: Taoussa planning authority.

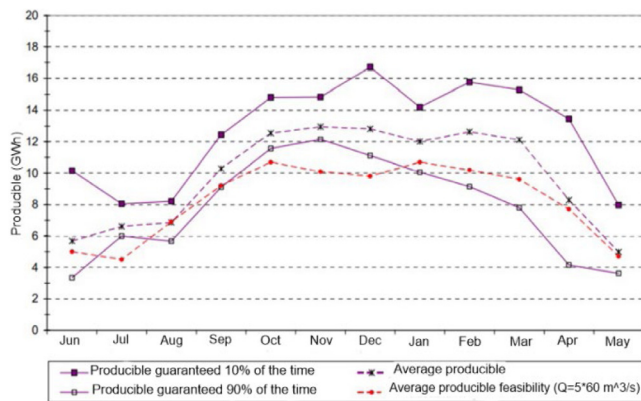


Fig. 6. Monthly producible (GWh).
Source: Taoussa planning authority.

energy analysis, and forward-looking development of scenarios. MESSAGE minimizes the total system cost under a set of constraints that concern the satisfaction of MAED's projected useful energy demand, equipment capacity, energy flow balances, resource availability, limits on the import and export of energy forms, environmental regulations, capacity and activity bounds, etc. It provides information on the primary and final energy mixes, and resource use; it establishes a capacity expansion plan for production units, a plan for exporting and importing different forms of energy, a forecast of investment needs in the energy system, an assessment of polluting emissions and waste, etc.

In this system modeling approach, an interactive systems procedure is carried out, where the feedback between models allows the verification of the consistency of MESSAGE and MAED equilibrium (de Lucena et al., 2010).

The MESSAGE model is designed as a linear and mixed-integer programming (LP & MIP) model. In LP applications, the objective function and constraints are linear functions of the decision variables. The MIP used in the MESSAGE modeling approach is due to integer values in the requirements of the best possible solution by certain decision variables. The MESSAGE model is a dynamic LP model of the global energy system. The objective function usually applied is to minimize the total costs of energy supply from the beginning to the end planning horizon. The process of finding the minimum value in MESSAGE is expressed by Eq. (1) (Nyasapoh, 2018).

$$\min \sum_q \sum_{t=1}^{nT} \left[d_t^o \Delta_t X_{qt} * i_{qt} + d_t^c \Delta_t Y_{qt} * O_{qt} \right] \quad (1)$$

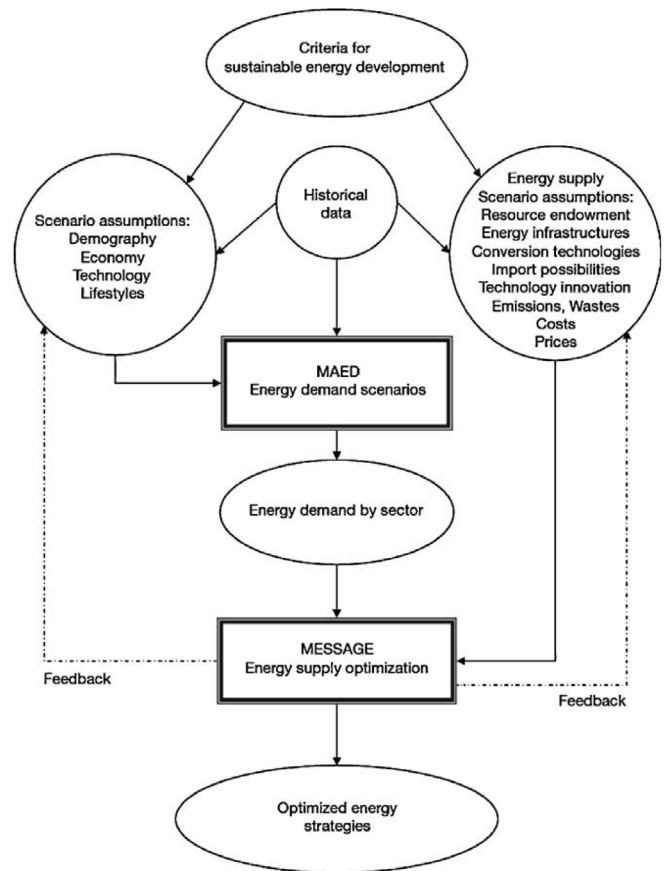


Fig. 7. Conceptual MAED-MESSAGE framework (de Lucena et al., 2010).

Eq. (2) is the equilibrium constraint between the electricity supply and demand of the objective function (Nyasapoh, 2018).

$$\sum_{q \in Pe} N_{qt} X_{qt} \geq \sum_{q \in Ce} X_{qt} + D_{et} \quad (2)$$

3.1. Program flow in MESSAGE

The model consists of the following components (IAEA, 2007):

- **MESSAGE data storage**—files to store the input data (preparation of the database: technology database (tdb), application database (adb), local database (ldb) and update database (upd));
- **MXG (Matrix Generation)**—a matrix generation program uses the database to generate a matrix of the model which is solved by the select solver in the optimization program;
- **OPT (Optimization Program)**—a set of solvers which can be used to solve the matrix generated; such as oslv2, glpk, cplex and mosek;
- **CAP (Calculation Program)**—a program for the post-processing of the solution for extraction of selected parts of the solution (generate information for tables and graphs).

After successful execution of MXG, OPT or CAP, MESSAGE displays the name of the program besides the title of “Done:” i.e. Done:MXG, Done:OPT and Done:CAP. Fig. 8 shows the flow of control and information between these components in the execution of the MESSAGE program.

The MESSAGE program in the case study of Taoussa is given in Appendix A.

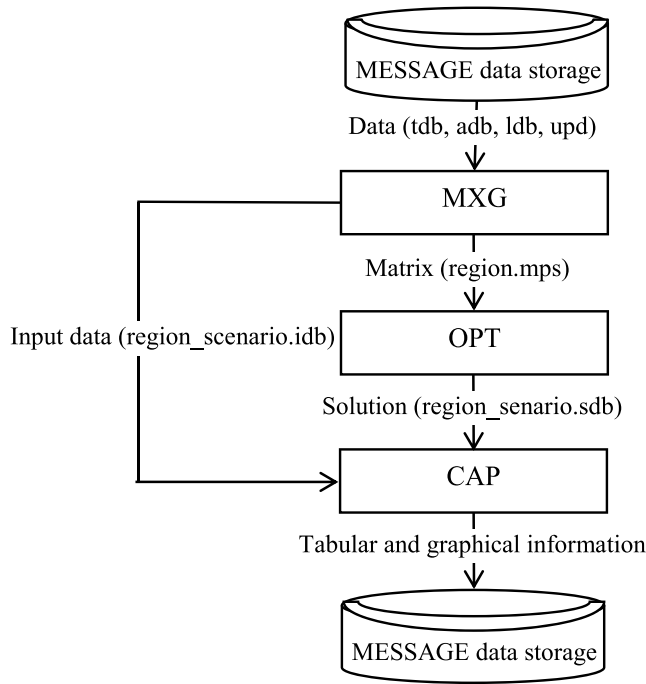


Fig. 8. Interrelationship between MESSAGE components (IAEA, 2007).

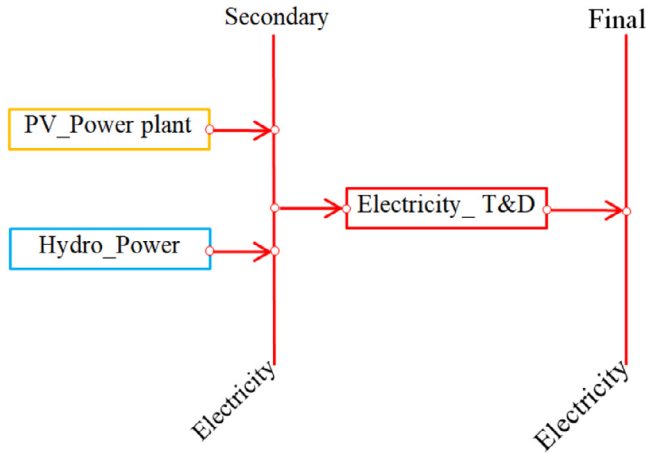


Fig. 9. Schematic flow diagram of a simple electricity system chains.

4. Background information for modeling a simple energy supply system

4.1. Electricity system structure

Fig. 9 presents the descriptive representation of simple electricity system chains in the MESSAGE model for the Taoussa area from supply to demand.

In this case study, the electricity lost during transmission & distribution between the point of generation and the point of delivery is given by Eq. (3).

$$E_{lost} = EP_t - ED_t \quad (3)$$

4.2. Electricity demand projection

Kanté et al. (2021) used the MAED model to forecast the electricity requirements for the Taoussa area in the north of Mali

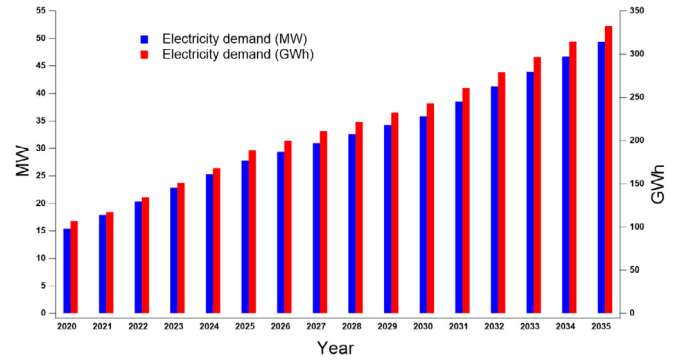


Fig. 10. Final electricity demand for the low scenario.

by three scenarios (LS, RS and HS) from 2020 to 2035. The construction of the divergent scenarios will make possible changes in the country's energy demand and the role that electricity could play in meeting that demand.

The basic elements of the energy demand scenario are:

- **Demography**
 - Population growth rate, rural/urban, labor force;
 - Accommodation: number, type, area...
- **Economy**
 - Different economic growth rates focus on industry/agriculture.

Other elements

- Changes in demographic factors such as urbanization, household size etc...;
- Macroeconomic structural changes;
- Changes in the industrial structure;
- Structural changes in the transportation system;
- Changes in lifestyles; e.g., use of cars or air conditioners;
- Changes in energy intensities of various economic activities due to technological development;
- Efficiency improvements.

The basic data used in the MAED model by scenario is mentioned in Table 3.

4.2.1. Final electricity demand by scenario

Figs. 10, 11 and 12 show the projection of total electricity requirements in the Taoussa area for the three scenarios over the period 2020–2035. Total electricity demand will rise by 2035 for the LS, RS and HS respectively 49.40 MW (332.57 GWh), 66.46 MW (472.61 GWh) and 89.47 MW (647.02 GWh); the average growth rate of electricity demand will be 8.13% for the LS, 10.31% for the RS and 12.56% for the HS. The annual electricity requirements by scenario, compared to the year 2020, will triple for LS while it will increase fourfold for RS and fivefold for HS. In addition, the distribution of electricity demand by sector is 56.77% in industry, 25.60% in household and 17.63% in service for LS, 56.13% in industry, 28.55% in household and 15.31% in service for RS and 61.16% in industry, 25.67% in household and 13.17% in service for HS, respectively (Kanté et al., 2021).

4.3. Load regions (LR)

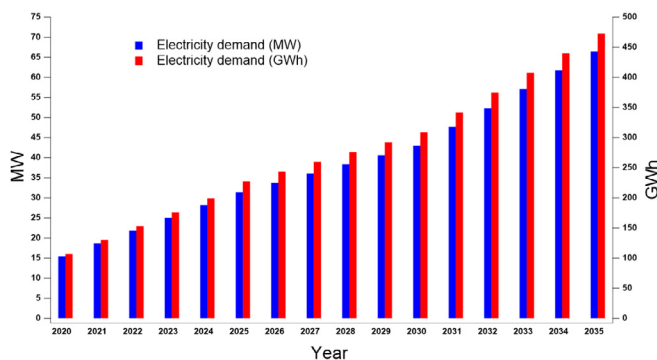
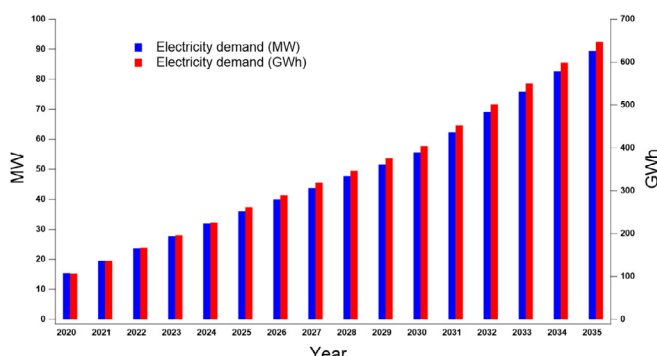
The structure of the load regions data (number of seasons, number of types of days for a season, number of parts of a day type) can be mathematically expressed by Eq. (4).

$$\sum_j \sum_i W_{ij} * n_j = W_k \quad (4)$$

Table 3

Evolution of the basic elements of the MAED model in the Taoussa area (Kanté et al., 2021).

Item	Unit	2020	2025	2030	2035
Socioeconomic					
Population	Thousand	754.07	895.60	1089.64	1351.4
Population growth rate	% per annum	–	3.50	4.00	4.40
Urban population ratio	%	42.00	44.00	46.00	48.00
Rural population ratio	%	58.00	56.00	54.00	52.00
GDP					
LS	USD Billion	1.30	1.44	1.62	1.84
RS		1.30	1.55	1.92	2.41
HS		1.30	1.75	2.39	3.27
GDP Growth rate					
LS	% p.a.	–	2.00	2.50	2.50
RS		–	3.60	4.40	4.60
HS		–	6.10	6.50	6.40
GDP per capita					
LS	USD Billion/Cap	1.72	1.60	1.49	1.36
RS		1.72	1.73	1.77	1.78
HS		1.72	1.95	2.20	2.42
Electricity penetration (rural & urban)					
LS	%	41.00	45.00	51.50	60.00
RS		41.00	58.50	70.00	77.50
HS		41.00	62.50	75.00	82.50
Specific electricity requirements (rural & urban)					
LS	kWh/dw/year	2894.47	2996.57	3102.49	3212.40
RS		2894.47	3077.35	3273.2	3483.04
HS		2894.47	3222.08	3588.17	3997.46

**Fig. 11.** Final electricity demand for the reference scenario.**Fig. 12.** Final electricity demand for the high scenario.

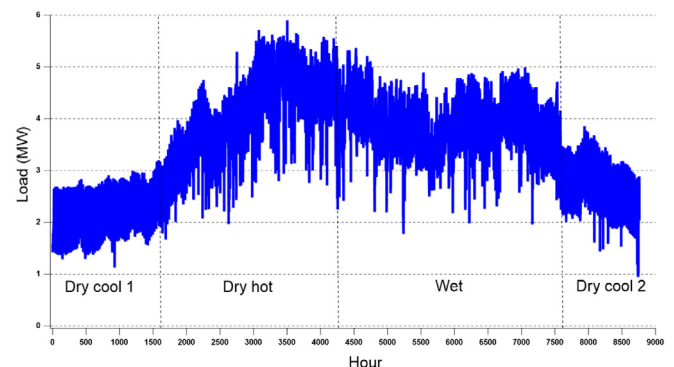
4.3.1. Seasons and types of days

In Table 4, the electricity demand system load is usually divided into 4 seasons (Dry hot, Wet, Dry cool 1&2), in the Taoussa

Table 4

Number of seasons, stating the date and alternative day types for seasons.

Season	Starting date	Type of days
Dry cool 1	1 January	Working days
Dry hot	11 March	Friday
Wet	1 July	Saturday/Sunday/Holiday
Dry cool 2	18 November	

**Fig. 13.** Hourly load curve (2019) (Kanté et al., 2021).

area there are 3 types of the day (workdays, Fridays and weekends), workdays include 4 days (Monday to Thursday) and the weekend (Saturday and Sunday) & holiday.

The data in Fig. 13 can be shown in detail in Fig. 14, Tables 5 and 6.

As shown in Fig. 14, the duration of daily time slices as a fraction from 24 h for each type of typical day (workday, Friday and weekend) by seasons (dry cool 1&2, dry hot and wet) selected. Assuming that the consumption of electricity is not uniform during 24 h of daily load curve by typical days in each season. So now 14 load regions can be divided for dry cool 1 (5 regions for workday, 5 regions for Friday and 4 regions for weekend). The other three seasons are similar, and there are a

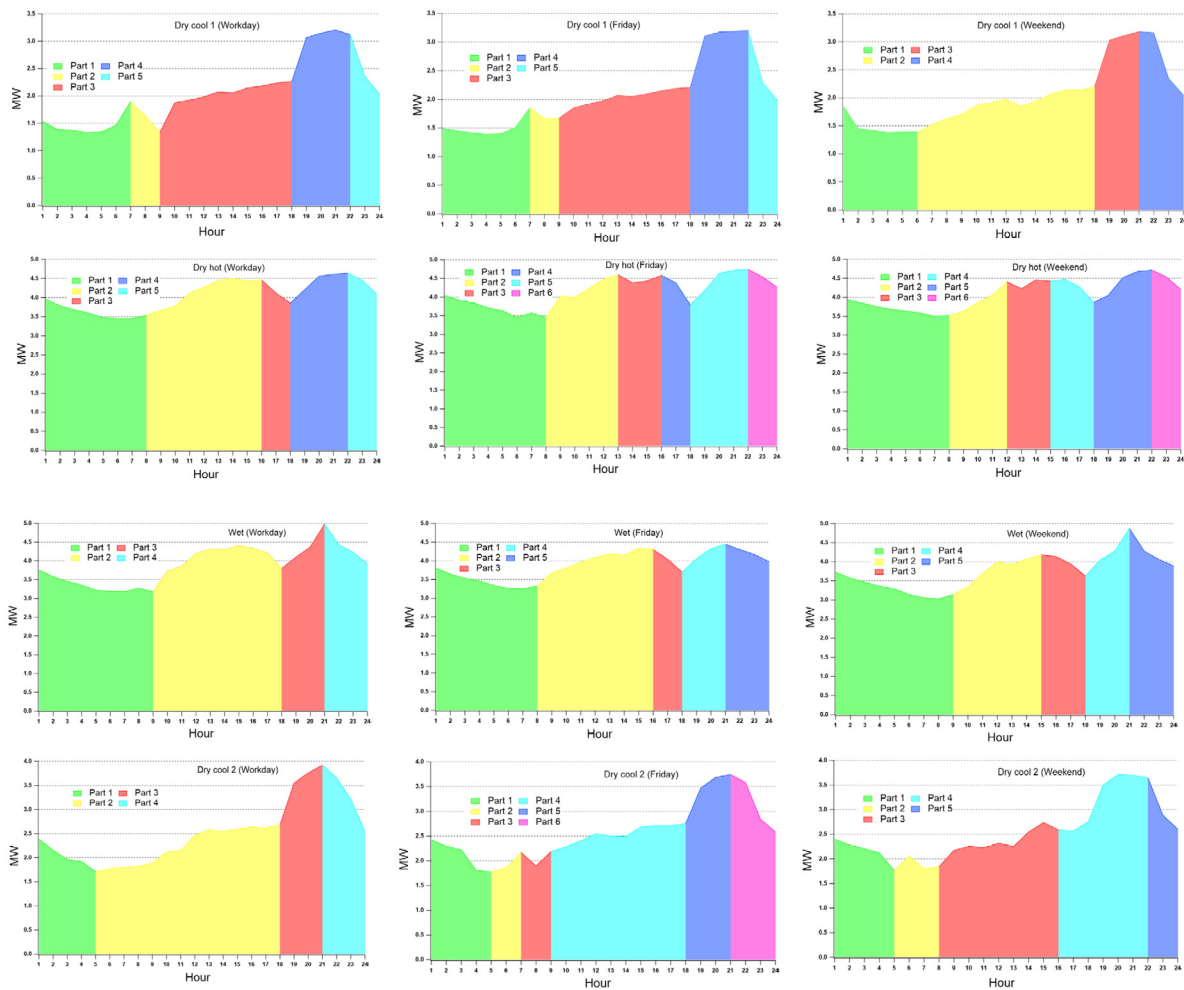


Fig. 14. Division in each typical day by season.

Table 5

Time distribution of 2019 to model seasonal variations in electricity demand.

Season	Duration	Day type	Length of load regions, fraction					
			1	2	3	4	5	6
Dry cool 1	January 01 to March 11	Workday	0.250	0.083	0.375	0.167	0.125	–
		Friday	0.250	0.083	0.375	0.167	0.125	–
		Weekend	0.208	0.500	0.125	0.167	–	–
Dry hot	March 12 to July 01	Workday	0.292	0.333	0.083	0.167	0.125	–
		Friday	0.292	0.208	0.125	0.083	0.167	0.125
		Weekend	0.292	0.167	0.125	0.125	0.167	0.125
Wet	July 02 to November 18	Workday	0.333	0.375	0.125	0.167	–	–
		Friday	0.292	0.333	0.083	0.125	0.167	–
		Weekend	0.333	0.250	0.125	0.125	0.167	–
Dry cool 2	November 19 to December 31	Workday	0.167	0.542	0.125	0.167	–	–
		Friday	0.167	0.083	0.083	0.375	0.125	0.167
		Weekend	0.167	0.125	0.333	0.250	0.125	–

total of 60 load regions in a year. Regarding the length of these load regions, MESSAGE calculates the number of days in a season for each day type from the duration for each season; see from fourth to ninth columns in Table 5. For example, if Monday is included in Workday then all Mondays in that season will be counted to compute the number of Workdays in that season.

Eq. (5) and Fig. 14 are used to elaborate Table 5.

$$L_{LRF} = \frac{H_{ijk}}{24} \quad (5)$$

Table 6 proposes a distribution pattern of the average annual electricity demand by the 60 load regions discussed above. The

first three columns of Table 6 include seasons, duration and their shares in the total annual demand, respectively; the following columns include day types for each season, the share of each day type and the share of each party in the electricity demand of a typical day, respectively. To further understand these data, take the data of dry cool 1-weekday as an example, 11.3% of the annual electricity demand will be in dry cool 1; 57.2% of the demand during a typical week of this season is on the working days of the week; and from the demand during a typical working day, 17.2% will be in region 1, 7.2% in region 2, 36.4% region 3, 23.8%

Table 6

Distribution of annual electricity demand by load regions.

Season	Duration	Shares by season	Day type	Shares by day type	Shares by parts of the day					
					1	2	3	4	5	6
Dry cool 1	January 01 to March 11	0.113	Workday	0.572	0.172	0.072	0.364	0.238	0.153	–
			Friday	0.144	0.176	0.071	0.364	0.237	0.152	–
			Weekend	0.284	0.154	0.455	0.171	0.220	–	–
Dry hot	March 12 to July 01	0.366	Workday	0.568	0.261	0.337	0.088	0.177	0.136	–
			Friday	0.146	0.263	0.203	0.135	0.090	0.174	0.136
			Weekend	0.286	0.265	0.154	0.134	0.135	0.175	0.138
Wet	July 02 to November 18	0.432	Workday	0.577	0.289	0.391	0.131	0.188	–	–
			Friday	0.144	0.261	0.339	0.090	0.130	0.181	–
			Weekend	0.279	0.296	0.246	0.136	0.133	0.190	–
Dry cool 2	November 19 to December 31	0.089	Workday	0.569	0.139	0.475	0.165	0.221	–	–
			Friday	0.145	0.142	0.059	0.066	0.365	0.161	0.207
			Weekend	0.287	0.148	0.092	0.301	0.309	0.150	–

Table 7

Techno-economic parameters of the power system and electricity-T&D.

Source: National Energy Authority of Mali.

	Investment costs (US \$/kWYr)	Fix O & M costs (US \$/kWYr)	Variable O & M costs (US \$/kWYr)	Efficiency (%)	Plant factor (%)	Plant life (yr)
Hydro	2752.00	44.03	763.48	55	95	50
PV	1005.00	11.06	1298.90	25	95	30
Electricity-T&D	–	–	142.35	85	98	60

region 4 and remaining 15.3% in region 5. The distribution pattern of the demand for the other three seasons are shown similarly.

Eqs. (6)–(8) are some simple utilities to help calculate hourly electricity load data into the seasonal representation used in MESSAGE.

$$SS = \frac{\sum_{D=1}^{ND_k} HA_D}{\sum_{k=1}^4 HA_k} \quad (6)$$

$$STD = \frac{\sum_{D=1}^{ND_j} HT_D}{\sum_{D=1}^7 HD_D} \quad (7)$$

$$SPD = \frac{\sum_{N_H=1}^{ND_H} HP_{N_H}}{\sum_{N_H=1}^{24} HD_{N_H}} \quad (8)$$

4.4. Techno-economic parameters

In the MESSAGE model for the Taoussa area, economic and technical input data for the conversion technologies (power systems) are shown in Table 7.

Cost savings is a term used to refer to the gains obtained from an economic process or activity. In this case study, it is the monetary amount resulting from the difference between the variable O&M costs (Hydro & PV) and initial investment costs during a given period. Eq. (9) shows the method for calculating this cost.

$$CS = \sum_{q=1}^2 \sum_{t=1}^{16} VC_{qt} - \left(\sum_{q=1}^2 \sum_{t=1}^{16} IC_{qt} + \sum_{q=1}^2 \sum_{t=1}^{16} FC_{qt} + \sum_{t=1}^{16} VC_{T\&D_t} \right) \quad (9)$$

where

$q = 1, 2$ (type of technology);
 $t = 1, \dots, 16$ (time horizon).

5. Optimization results and interpretation

This study applies sustainable electricity system development approaches to the model to integrate into the future final electricity supply-mix options in meeting Taoussa's long-term electricity

demand forecasting from 2020 to 2035 to find the reasonable cost for electricity supply and mitigate greenhouse gas (GHG) emissions. This chapter introduces MESSAGE model results and discussions developed on the equilibrium between electricity demand & supply and local CO₂ emissions.

5.1. Electricity production capacity by technology

The optimal mix of power generation (hydroelectric and PV) in Taoussa for the period 2020–2035 have been done with the IAEA's model MESSAGE for three scenarios (LS, RS and HS). As shown in Table 8 and Fig. 15, average annual power generation increases year by year, from 15.72 MW (109.02 GWh) (15.37 MW (106.58 GWh) for hydroelectricity and 0.35 MW (2.43 GWh) for PV) in 2020 for all scenarios to 50.41 MW (339.36 GWh) (17.10 MW (115.13 GWh) for hydroelectricity and 33.31 MW (224.25 GWh) for PV), 67.82 MW (482.25 GWh) (17.10 MW (121.60 GWh) for hydroelectricity and 49.66 MW (360.65 GWh) for PV) and 91.30 MW (660.25 GWh) (17.10 MW (123.67 GWh) for hydroelectricity and 74.20 MW (536.58 GWh) for PV) in 2035 for the Low scenario, reference scenario and High scenario respectively. The annual increase of power generation in LS is equivalent to 8.13%, while RS and HS projection increases are 10.32% and 12.56% respectively. In conformity with the forecast electricity demand, the two different power plants (hydroelectric and photovoltaic) are requested for generating electricity starting with the year 2020 in all scenarios. The largest share of future total electricity production capacity will have to be based on the PV power plant, namely about 49% for the Low Scenario, 57% for the Reference Scenario and about 66% for the High Scenario.

5.2. Comparative evolution between electricity demand and production

In Table 9, the energy loss between electricity demand and production (transmission & distribution) is from 0.31 MW (2.17 GWh) in 2020 for all scenarios to 1.01 MW (6.79 GWh) by 2035 for the low scenario, 1.36 MW (9.64 GWh) by 2035 for reference scenario and 1.83 MW (13.23 GWh) by 2035 for the high scenario. It is worth noting that the generation of electricity will be around 1.02 times of electricity demand during 2020–2035 in the Taoussa area.

Table 8
Evolution of electricity production and Production shares by technology.

Year	Low scenario					Reference scenario					High scenario				
	Production		Increase in MW	Shares by technology (%)		Production		Increase in MW	Shares by technology (%)		Production		Increase in MW	Shares by technology (%)	
	GWh	MW		Hydro	PV	GWh	MW		Hydro	PV	GWh	MW		Hydro	PV
2020	109.01	15.72	–	97.84	2.16	109.01	15.72	–	97.84	2.16	109.01	15.72	–	97.84	2.16
2021	119.67	18.24	16.03	88.86	11.14	132.65	19.00	20.83	86.19	13.81	139.44	19.93	26.74	83.01	16.99
2022	136.91	20.77	13.81	79.01	20.99	156.23	22.27	17.19	74.71	25.29	169.88	24.13	21.10	69.99	30.01
2023	154.15	23.29	12.14	70.80	29.20	179.80	25.53	14.67	65.65	34.35	200.31	28.34	17.42	60.11	39.89
2024	171.39	25.81	10.82	63.91	36.09	203.45	28.81	12.83	58.12	41.88	230.75	32.54	14.84	52.03	47.97
2025	192.86	28.33	9.77	59.04	40.96	232.09	32.07	11.34	52.76	47.24	266.72	36.74	12.92	46.31	53.69
2026	203.90	29.98	5.84	56.16	43.84	248.74	34.43	7.35	49.30	50.70	295.77	40.72	10.83	41.95	58.05
2027	214.94	31.62	5.48	53.46	46.54	265.30	36.78	6.82	46.27	53.73	324.91	44.71	9.80	38.24	61.76
2028	225.97	33.28	5.23	50.94	49.06	281.94	39.13	6.41	43.60	56.40	354.03	48.70	8.92	35.11	64.89
2029	237.01	34.92	4.94	48.63	51.37	298.51	41.48	6.00	41.21	58.79	383.08	52.68	8.17	32.46	67.54
2030	248.05	36.57	4.73	46.52	53.48	315.15	43.84	5.68	39.01	60.99	412.22	56.67	7.57	30.17	69.83
2031	266.31	39.34	7.56	43.38	56.62	348.57	48.63	10.94	35.16	64.84	461.85	63.60	12.23	26.89	73.11
2032	284.57	42.10	7.03	40.62	59.38	381.98	53.43	9.86	32.01	67.99	511.41	70.52	10.88	24.25	75.75
2033	302.84	44.87	6.57	38.11	61.89	415.41	58.22	8.98	29.37	70.63	561.05	77.45	9.82	22.08	77.92
2034	321.09	47.64	6.19	35.89	64.11	448.83	63.02	8.24	27.13	72.87	610.61	84.37	8.93	20.27	79.73
2035	339.36	50.41	5.80	33.92	66.08	482.25	67.82	7.61	25.22	74.78	660.25	91.30	8.21	18.73	81.27

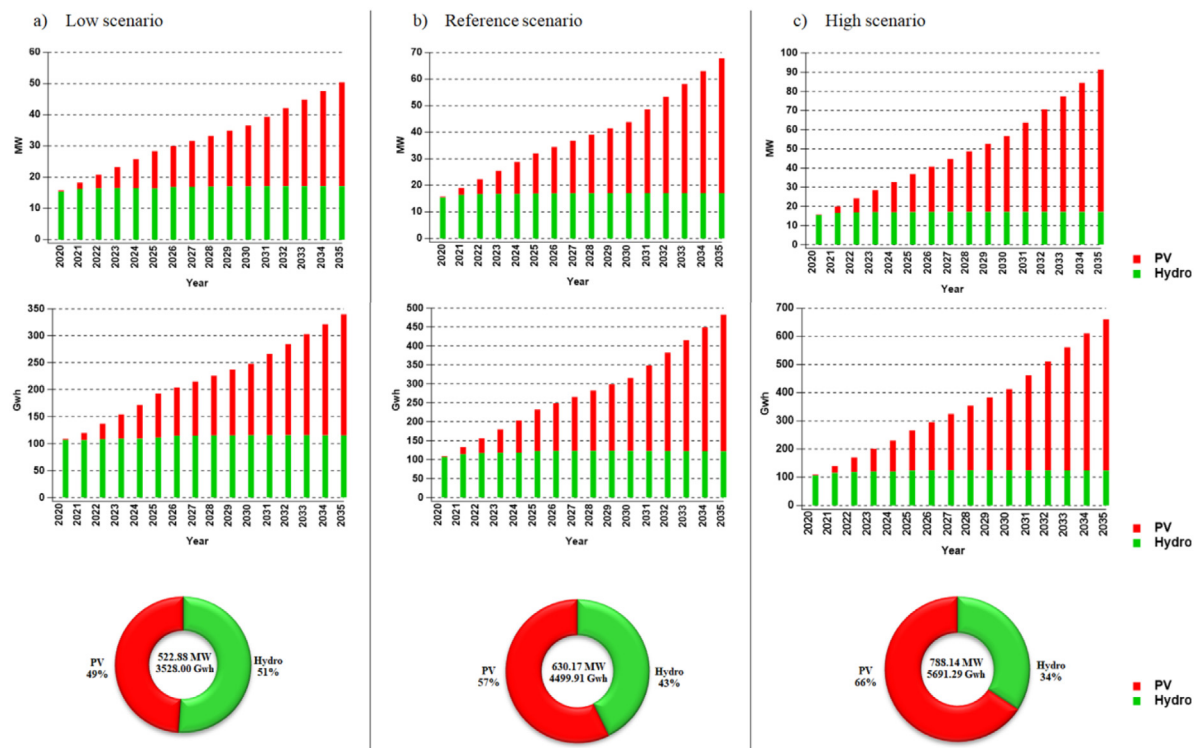


Fig. 15. Evolution of electricity production capacity by technology.

5.3. Maximum capacity by technology during the planning period

In Appendix B, the maximum hydropower is 18.60 MW from April to May 2020 for all scenarios. In this period, the total electricity generation is 20.78 MW (89.51% hydropower and 10.49% PV). It is worth noting that hydropower will become completely saturated. The maximum power of PV for the three scenarios is 49.72 MW, 72.79 MW and 103.92 MW from April to May 2035 respectively for the low scenario, reference scenario and high scenario. In those periods, the maximum share of hydropower and PV in total electricity generation of this period will be 66.61 MW (74.33% PV and 25.67% hydropower) for LS, 89.61 MW (80.92% PV and 19.08% hydropower) for RS and 120.63 MW (85.83% PV and 14.17% hydropower) for HS.

5.4. Power costs

Tables 10–12 give the future evolution of annual investment costs, fix O&M costs and variable O&M costs by scenario. The investment cost of hydroelectric power will start at 2025 (2248.93 US\$/kWyr) for LS, 2024 (1149.24 US\$/kWyr) for RS and 2023 (49.54 US\$/kWyr) for HS and then stabilize at 1099.70 US\$/kWyr in 2026 for LS, 2025 for RS and 2024 for HS respectively. The investment cost of PV will continue to vary rapidly over time by 2034 to reach 4869.53 US\$/kWyr for LS, 7373.40 US\$/kWyr for RS and 13 713.43 US\$/kWyr for HS respectively.

The fix O&M cost of hydroelectric power will vary from 598.24 US\$/kWyr in 2020 for the three scenarios to 774.19 US\$/kWyr in 2025 for LS, 791.78 US\$/kWyr in 2024 for RS and 809.38 US\$/kWyr in 2023 for HS respectively. It starts to stabilize from

Table 9
Losses in transmission & Distribution by scenario.

Year	Low scenario		Reference scenario		High scenario	
	GWh	MW	GWh	MW	GWh	MW
2020	2.17	0.31	2.17	0.31	2.17	0.31
2021	2.39	0.36	2.68	0.38	2.79	0.40
2022	2.74	0.42	3.12	0.44	3.41	0.48
2023	3.08	0.47	3.56	0.51	4.02	0.57
2024	3.43	0.52	4.07	0.58	4.64	0.65
2025	3.86	0.57	4.61	0.64	5.37	0.74
2026	4.08	0.60	4.99	0.69	5.90	0.81
2027	4.30	0.63	5.28	0.73	6.51	0.90
2028	4.52	0.67	5.66	0.79	7.11	0.98
2029	4.74	0.70	5.96	0.83	7.64	1.05
2030	4.96	0.73	6.33	0.88	8.25	1.13
2031	5.33	0.79	6.99	0.98	9.27	1.28
2032	5.69	0.84	7.65	1.07	10.22	1.41
2033	6.06	0.90	8.32	1.17	11.25	1.55
2034	6.42	0.95	8.98	1.26	12.20	1.69
2035	6.79	1.01	9.64	1.36	13.23	1.83

2026 (792.58 US\$/kWyr) for LS, 2025 (792.58 US\$/kWyr) for RS and 2024 (792.58 US\$/kWyr) for HS respectively. Between 2020 and 2035, the fix O&M cost of PV will be multiplied by 13.04 times for LS, 8.19 times for RS and 11.69 times for HS respectively.

The variable O&M cost of hydroelectric power starts to grow from 12 386.40 US\$/kWyr in 2021 to 13 054.59 US\$/kWyr in 2032 for LS, from 12 503.96 US\$/kWyr in 2021 to 13 049.37 US\$/kWyr in 2029 for RS and from 12 630.56 US\$/kWyr in 2021 to 13 042.49 US\$/kWyr in 2026 for HS respectively. It will be almost constant at 13 055.50 US\$/kWyr in 2033 for LS, 2030 for RS and 2027 for HS respectively. While the variable O&M cost of PV will evolve in an increasing way for all scenarios, the increase will be from 2169.76 US\$/kWyr in 2021 to 3591.83 US\$/kWyr in 2035 for LS, from 2950.56 US\$/kWyr in 2021 to 6229.40 US\$/kWyr in 2035 for RS and from 3941.30 US\$/kWyr in 2021 to 8999.49 US\$/kWyr in 2035 for LS respectively. Between 2020 and 2035, the variable O&M cost of T&D will increase by 3.21 times for LS, 4.31 times for RS and 5.81 times for HS respectively.

In the PV, hydroelectric power and electricity transmission & distribution, the cumulative annual investment and O&M costs for the period of 2020–2035 will be 688 553.46 US\$/kWyr (8.79% for investment cost, 2.56% for fix O&M costs and 88.65% for variable O&M costs) for the low scenario, 862 756.63 US\$/kWyr (9.21% for investment cost, 2.24% for fix O&M costs and 88.55% for variable O&M costs) for the reference scenario and 1 160 422.58 US\$/kWyr (12.48% for investment cost, 2.13% for fix O&M costs and 85.40% for variable O&M costs) for the high scenario, the cost could be saved by scenarios will be 383 409.68 US\$/kWyr for LS, 485 733.25 US\$/kWyr for RS and 597 118.9 US\$/kWyr for HS, respectively.

5.5. Environmental sustainability

Global CO₂ emissions from the electricity sector increased by 7% (compared to 2020). In total, 62% of the world's electricity production in 2021 was based on fossil fuels (36% for coal alone), 28% on renewable energies and around 10% on nuclear energy according to the IEA (CONNAISSANCE DES ÉNERGIES, 2022). The production of electricity from primary renewable resources (wind, sun, water, geothermal) is much more environmentally neutral and almost inexhaustible, as there is no combustion as with coal, fuel and gas. The combustion of diesel or gas oil emits on average 0.267 t CO₂/MWh (EUROPE, 2010), the use of hydroelectric and PV power plants in these large-scale scenario studies between 2020 and 2035 will avoid the emission of 941.98 kt of CO₂ for LS, 1201.48 kt for RS and 1519.58 kt for HS, respectively.

6. Conclusion and future work

Long-term forecasting potential of hydro and solar photovoltaic electricity generation in the Taoussa area using MESSAGE to obtain the equilibrium between sustainable electricity supply under electricity demand forecasted by three scenarios (LS, RS and HS).

According to the obtained long-term optimization results of a large-scale hybrid renewable electricity supply strategy from the case study, the annual increase of power generation during the study period 2020–2035 for LS, RS and HS are 8.13%, 10.32% and 12.56%, respectively. The maximum capacity of the hydropower plant is 18.60 MW from April to May 2020 for all scenarios; the PV power plant under three scenarios is 49.72 MW for LS, 72.79 MW for RS and 103.92 MW for LS from April to May 2035. The share of future total electricity production capacity of PV and hydropower in LS, RS and LS are 49% and 51%, 57% and 43%, 66% and 34%, respectively. In three scenarios, the electricity production using hydro and PV power plants will be around 1.02 times of electricity demand during the study period. The cumulative annual investment and O&M costs for this hybrid system will be 688 553.46 US\$/kWyr for the LS, 862 756.63 US\$/kWyr for the RS and 1 160 422.58 US\$/kWyr for the HS. For all scenarios, the electricity generated from the PV power supply will be the largest electricity producer in Taoussa.

This study has several benefits for the development of the Taoussa area in north Mali. These benefits include minimizing electricity supply costs, minimizing CO₂ emissions, minimizing reliance on imported energy, increasing local job and business opportunities and contributing to the local economy. Despite these benefits, renewable energy alternatives account for fluctuations in intermittent renewables supplying electricity. In the Taoussa area, the hydroelectric power is under 10 MW from April to August, the sunshine is fairly regular except for certain days of the rainy season. Most of the electricity generated during the day comes from solar energy, particularly from 6 a.m. to 5 p.m.

The future work will be based on energy storage technology to improve the stability and safety of the electricity supply during the intermittency and fluctuation times of the Taoussa area.

CRedit authorship contribution statement

Moussa Kanté: Conception and design, Final approval of the version to be published. **Shuai Deng:** Analysis and interpretation of data. **Yang Li:** Critically revising the article for important intellectual content. **Mansa Kanté:** Acquisition of data. **Souleymane Coulibaly:** Drafting the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Table 10
Investment requirements for the low scenario in US\$/kWyr.

Year	Investment costs		Fix O&M costs		Variable O&M costs		
	Hydro	PV	Hydro	PV	Hydro	PV	T&D
2020	0.00	0.00	598.24	44.18	11 737.48	455.69	2238.38
2021	0.00	0.00	844.57	198.79	12 386.40	2625.45	2597.16
2022	0.00	0.00	826.97	187.75	12 529.51	5655.74	2955.94
2023	0.00	2941.87	809.38	176.70	12 584.71	8835.59	3314.72
2024	0.00	4928.84	791.78	198.02	12 557.23	12 156.10	3673.50
2025	2248.93	2895.00	774.19	241.19	12 491.54	15 541.59	4032.28
2026	1099.70	3300.64	792.58	261.99	12 846.22	17 085.35	4267.60
2027	1099.70	3314.69	792.57	287.26	12 901.56	19 125.12	4501.45
2028	1099.70	3300.54	792.58	312.68	12 937.55	21 211.06	4736.77
2029	1099.70	3314.75	792.58	337.94	12 964.88	23 298.46	4970.63
2030	1099.70	4869.53	792.58	363.36	12 988.82	25 404.88	5205.94
2031	1099.70	4869.64	792.58	405.88	13 025.72	28 933.96	5599.58
2032	1099.70	4869.53	792.57	448.40	13 054.59	32 476.70	5993.22
2033	1099.70	4883.80	792.58	490.92	13 055.50	36 067.03	6386.86
2034	1099.70	4869.53	792.58	533.60	13 055.51	39 672.14	6781.97
2035	0.00	0.00	792.58	576.12	13 055.52	43 263.97	7175.61

Table 11
Investment requirements for reference scenario in US\$/kWyr.

Year	Investment costs		Fix O&M costs		Variable O&M costs		
	Hydro	PV	Hydro	PV	Hydro	PV	T&D
2020	0.00	0.00	598.24	103.08	11 737.48	455.69	2238.38
2021	0.00	0.00	844.57	206.15	12 503.96	3406.25	2704.65
2022	0.00	0.00	826.97	198.79	12 701.58	7311.34	3169.47
2023	0.00	5462.23	809.38	191.43	12 772.00	11 432.86	3634.28
2024	1149.24	5215.71	791.78	244.15	12 765.35	15 698.73	4100.55
2025	1099.70	3964.30	792.58	294.16	12 911.62	19 691.19	4565.36
2026	1099.70	3950.14	792.58	330.41	12 956.79	22 676.02	4900.91
2027	1099.70	3964.30	792.57	366.49	12 991.55	25 665.31	5234.99
2028	1099.70	3950.04	792.58	402.74	13 023.21	28 673.15	5570.54
2029	1099.70	3964.30	792.58	438.83	13 049.37	31 677.11	5904.62
2030	1099.70	7373.40	792.58	475.07	13 055.52	34 728.34	6240.15
2031	1099.70	7373.51	792.58	548.82	13 055.54	40 957.71	6922.86
2032	1099.70	7373.40	792.57	622.56	13 055.47	47 187.23	7605.56
2033	1099.70	7373.40	792.58	696.31	13 055.50	53 416.62	8288.26
2034	1099.70	7373.40	792.58	770.05	13 055.51	59 646.01	8970.95
2035	0.00	0.00	792.58	843.80	13 055.52	65 875.41	9653.65

Table 12
Investment requirements for the high scenario in US\$/kWyr.

Year	Investment costs		Fix O&M costs		Variable O&M costs		
	Hydro	PV	Hydro	PV	Hydro	PV	T&D
2020	0.00	0.00	598.24	136.51	11 737.48	455.69	2238.38
2021	0.00	0.00	844.57	273.03	12 630.56	4396.99	2836.84
2022	0.00	4946.78	826.97	263.27	12 890.29	9415.79	3435.29
2023	49.54	9177.42	809.38	307.94	12 974.27	14 733.61	4033.73
2024	1099.70	8669.51	792.58	399.14	12 922.07	20 283.09	4632.18
2025	1099.70	8253.90	792.58	484.75	12 991.18	25 626.19	5230.64
2026	1099.70	8272.94	792.58	565.79	13 042.49	30 707.98	5797.13
2027	1099.70	8272.79	792.57	647.05	13 055.47	35 868.23	6365.08
2028	1099.70	8253.90	792.58	728.30	13 055.50	41 050.57	6933.03
2029	1099.70	8272.79	792.58	809.34	13 055.51	46 219.64	7499.52
2030	1099.70	13 713.43	792.58	890.59	13 055.52	51 401.97	8067.47
2031	1099.70	13 694.66	792.58	1031.68	13 055.54	60 401.44	9053.74
2032	1099.70	13 713.43	792.57	1172.57	13 055.47	69 387.81	10 038.58
2033	1099.70	13 694.53	792.58	1313.67	13 055.50	78 387.31	11 024.87
2034	1099.70	13 713.43	792.58	1454.56	13 055.51	87 373.53	12 009.69
2035	0.00	0.00	792.58	1595.66	13 055.52	96 373.02	12 995.97

Research ethics

We further confirm that any aspect of the work covered in this manuscript that has involved human patients has been conducted with the ethical approval of all relevant bodies and that such approvals are acknowledged within the manuscript.

Appendix A

MESSAGE program of the Taoussa area

<https://github.com/moussankante/MESSAGE-program-of-the-Taoussa-area.git>

Appendix B

Total electricity production in the reference scenario, 2020–2035, by individual power plants (MW)



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